

DISTAF

**Digital Identity Systems
Trustworthiness Assessment
Framework**

User Manual

Version 1.0

Acknowledgements.

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The Institute is named in honour of Alan Turing, whose pioneering work in theoretical and applied mathematics, engineering and computing is considered to have laid the foundations for modern-day data science and artificial intelligence. It was established in 2015 by five founding universities and became the United Kingdom's (UK) National Institute for Data Science and Artificial Intelligence. Today, Turing brings together academics from 13 of the UK's leading universities and hosts visiting fellows and researchers from many international centres of academic excellence. Turing also liaises with public bodies and is supported by collaborations with influential organisations.

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1. Log-in and Log-out

The login process is a mandatory step to verify the identity of a user attempting to access the system. It involves the following steps:

1. Connect to the site at URL:

trustworthiness-framework-webapp.azurewebsites.net

2. Enter your username and password.
3. Press the **Log In** button.

Welcome! Please enter your user name and password below.

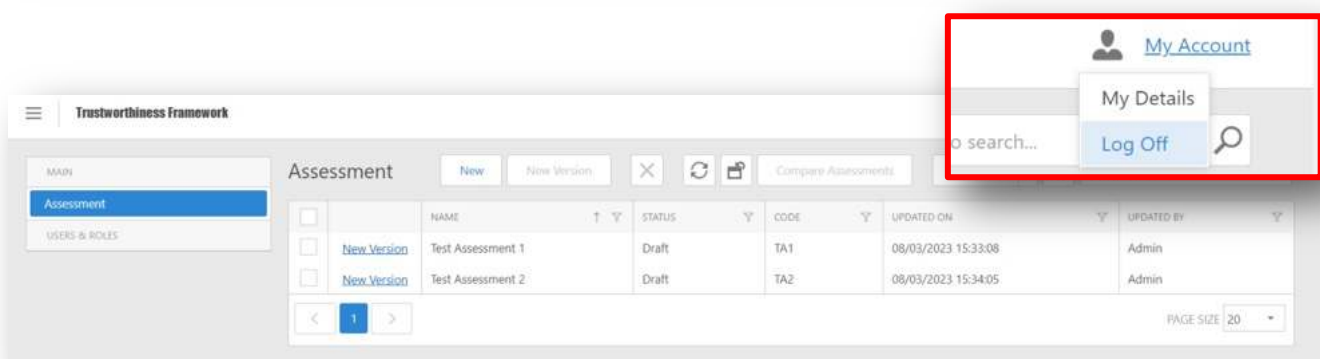
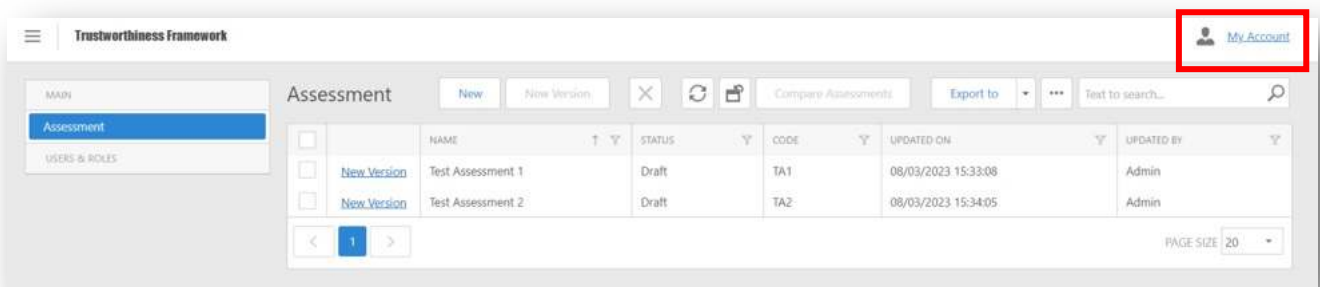
USER NAME:
Assessor

PASSWORD:

Log In

Symmetrically, the log-out process is used to close the current session, as follows:

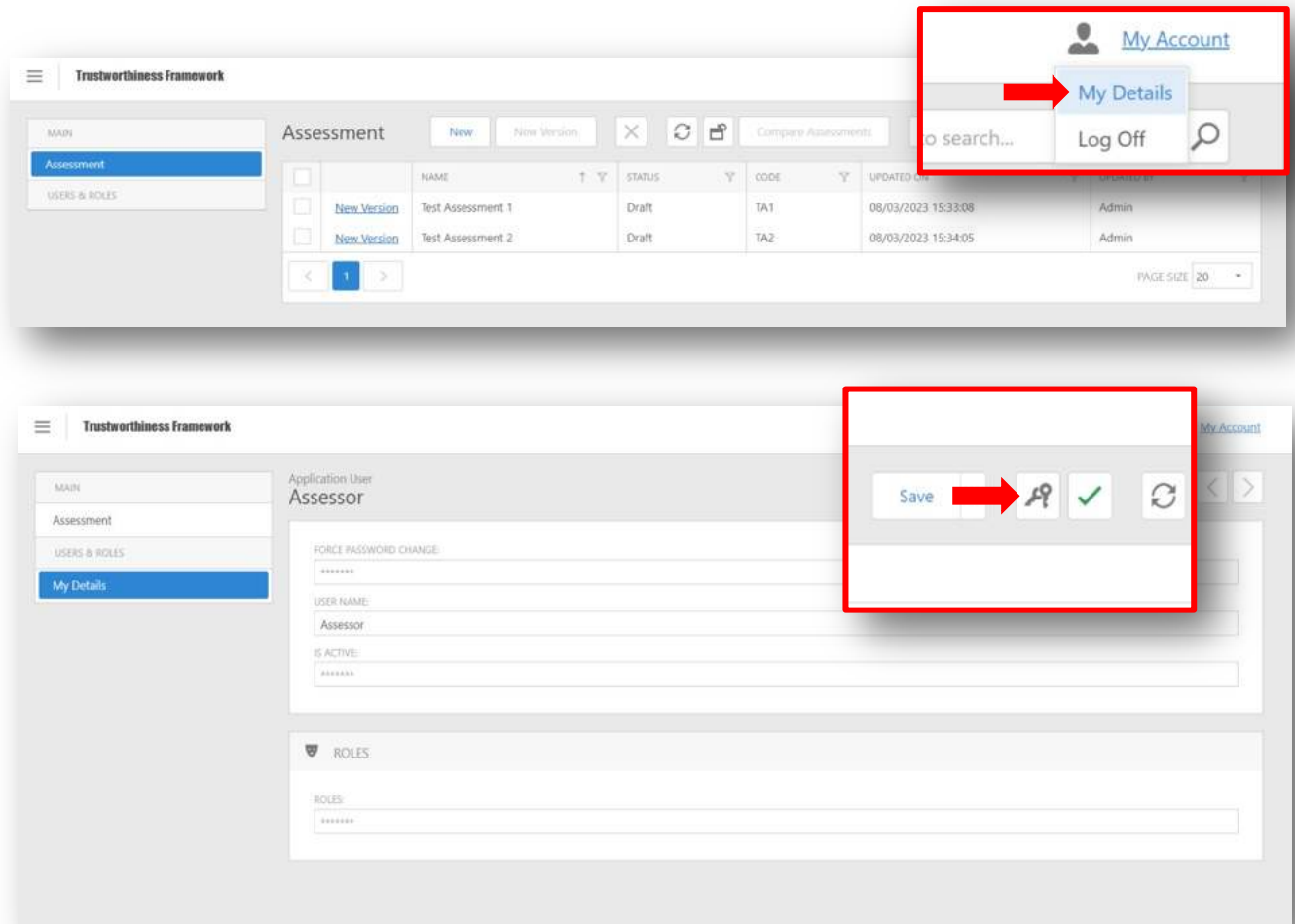
1. On the top right page select the **My Account** icon
2. Select the **Log Off** item from the drop-down menu.



2. How to change own password

Every user has the possibility of changing its own password, with the following steps:

1. On the top right page select the **My Account** icon
2. Select the **My Details** item from the drop-down menu.
3. Select the keys icon at the top right to access the **Change Password** page.



Once inside the Change Password dialog:

1. Type the old password in the appropriate field.
2. Enter the new password in the appropriate field.
3. Repeat the new password in the appropriate field.
4. Select the **Ok** button at the top right to confirm the change.
5. Select the **Cancel** button to undo the change and return to the **My Details** page.

Confirm
the change

The image shows a 'Change Password' dialog box. It has a title bar with 'Change Password' and two buttons: 'OK' and 'Cancel'. Below the title bar is a warning icon and a message: 'To change your password, please enter your old and new passwords in the text fields below.' There are three text input fields: 'OLD PASSWORD:', 'NEW PASSWORD:', and 'CONFIRM PASSWORD:'. Red arrows point to these fields with labels: 'Old Password' points to the first field, 'New Password' points to the second field, and 'New Password' points to the third field. Another red arrow points to the 'Cancel' button with the label 'Undo the change'.

3. Creating a new user and assigning roles

When logged-in as an administrator, it is possible to create a new user and to assign the corresponding permission (roles):

1. Select **Application User** from the menu on the left in the **User and Roles** section.
2. Select the **New** button at the top right.
3. Enter the name of the new user in the space provided.
4. Make sure the **Is Active** box is checked.

The image is a screenshot of the 'Trustworthiness Framework' application. On the left, there is a sidebar menu with 'Assessment' and 'USERS & ROLES'. Under 'USERS & ROLES', 'Application User' is selected and highlighted. In the main area, the title is 'Application User'. At the top right of this area, there is a 'New' button, which is highlighted with a red box. Below the title bar, there is a table with columns: 'USER NAME', 'FORCE PASSWORD CHANGE', and 'IS ACTIVE'. The table contains three rows: 'Admin', 'Assessor', and 'External'. The 'IS ACTIVE' column has checkboxes, all of which are checked. At the bottom right, there is a 'PAGE SIZE' dropdown set to '20'.

Now it is possible to assign roles, following the steps:

Trustworthiness Framework

Application User
John Smith

☐ Force Password Change

USER NAME:
John Smith

☒ Is Active

ROLES

New Link Unlink

NAME IS ADMINISTRATIVE CAN EDIT MODEL

No data to display

1. Click on the **Link** item in the **Roles** box below.
2. Select the role you want to assign to the user.
3. Select the blue "Ok" button at the bottom right to confirm the operation.

Trustworthiness Framework

Application User
John Smith

☐ Force Password Change

USER NAME:
John Smith

☒ Is Active

ROLES

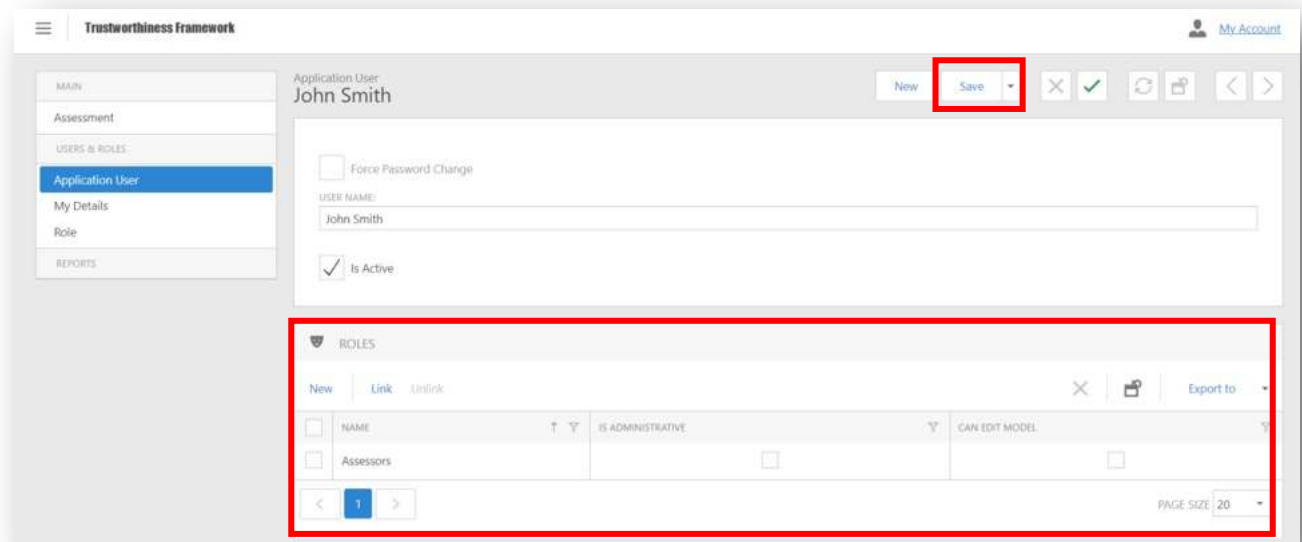
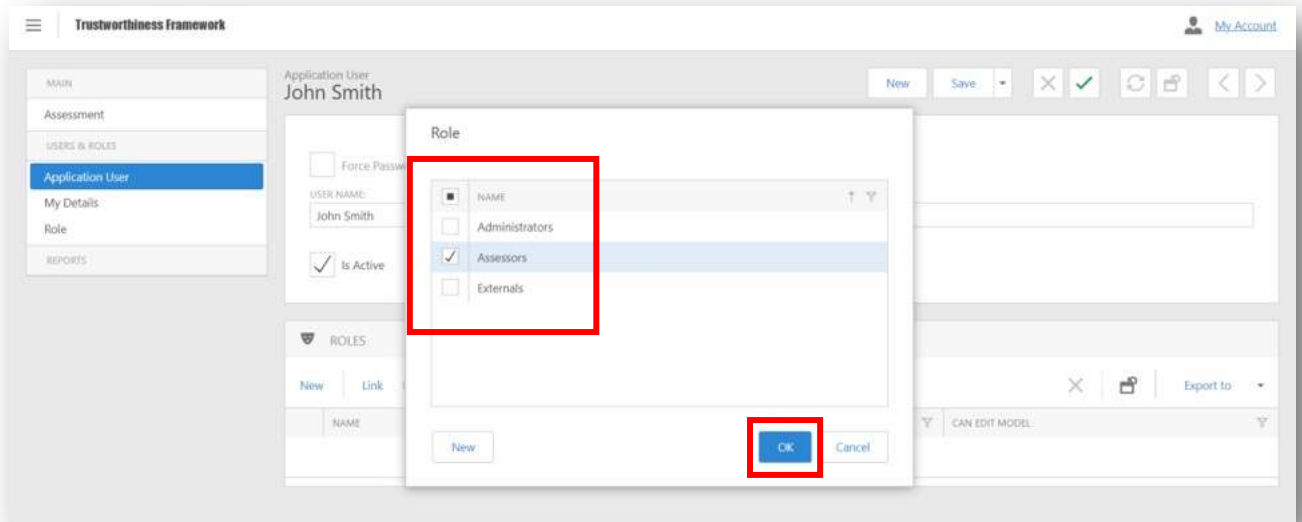
New Link Unlink

NAME IS ADMINISTRATIVE CAN EDIT MODEL

No data to display

4. In the "Role" pane you can now view the role assigned to the new user.

5. Press the "**Save**" button at the top right to save the changes.

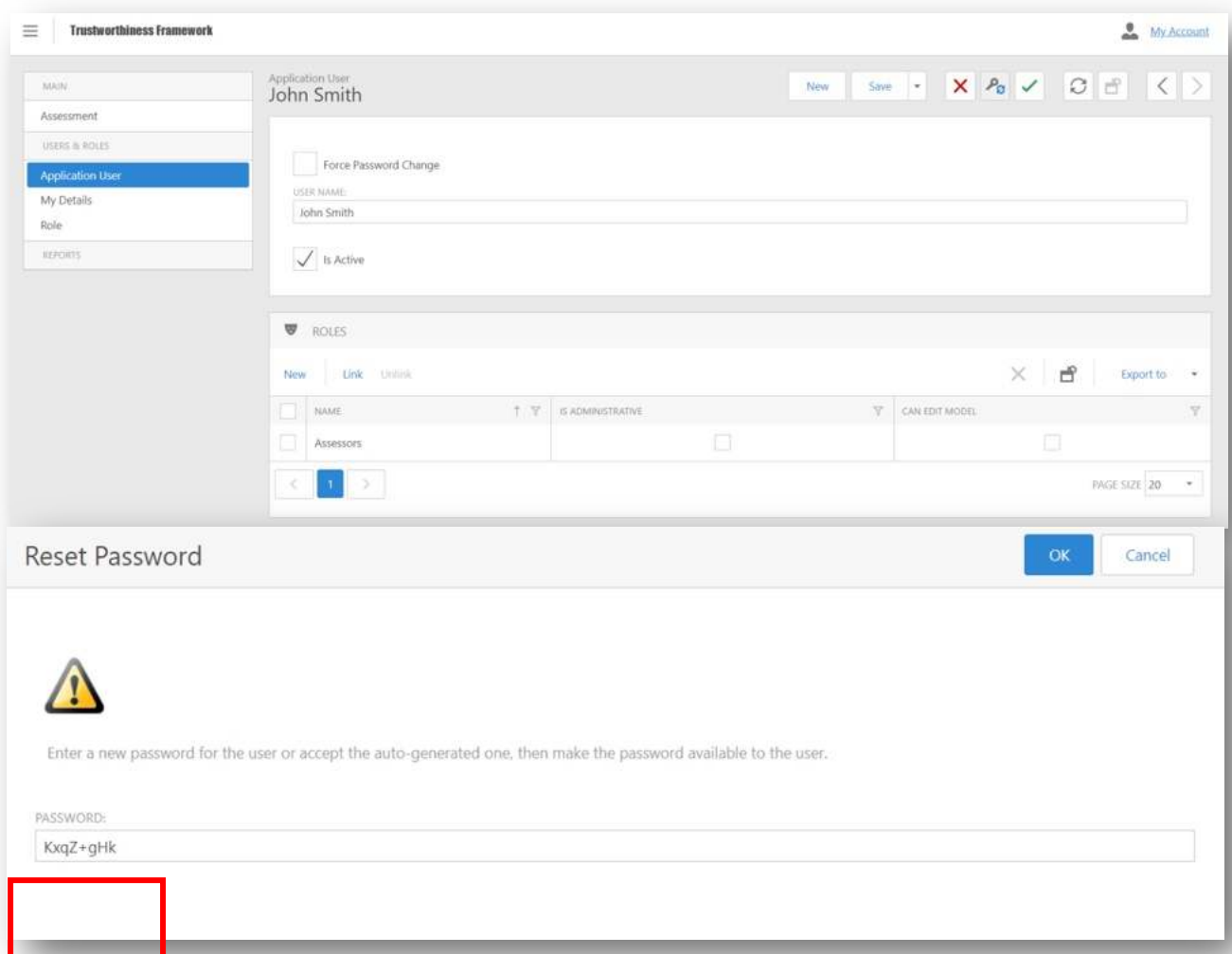
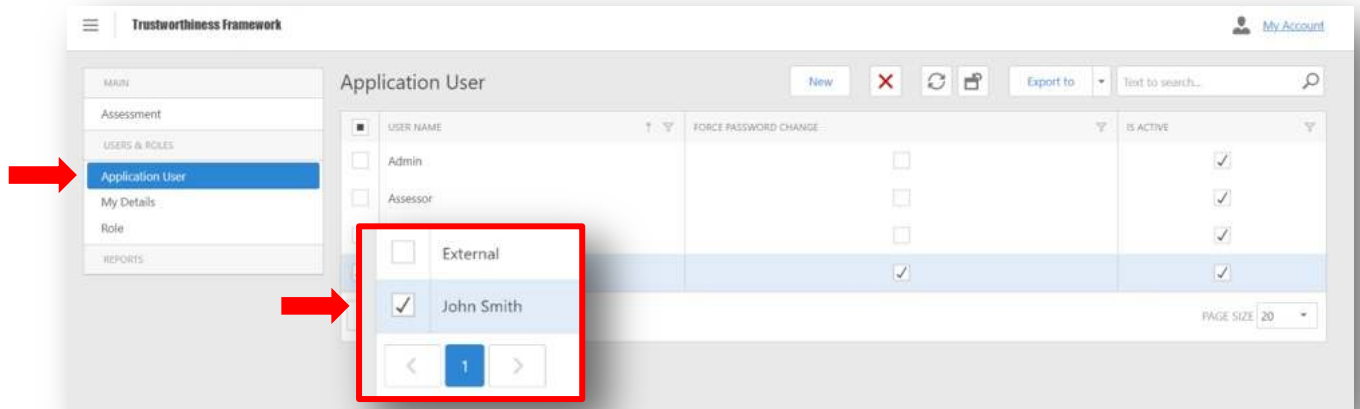


4. Resetting a user password

In case a user forgot their own password, it is possible to reset it with the following steps:

1. In the menu on the left select the page "**Application User**"
2. Check the relevant username.
3. Click on the user's name to open his page.
4. Click on the "key" symbol from the menu at the top right.

5. The system will automatically generate the new password that the user will have to insert at the next access.
6. Click the "OK" button at the top right to confirm the operation and return to the user page.

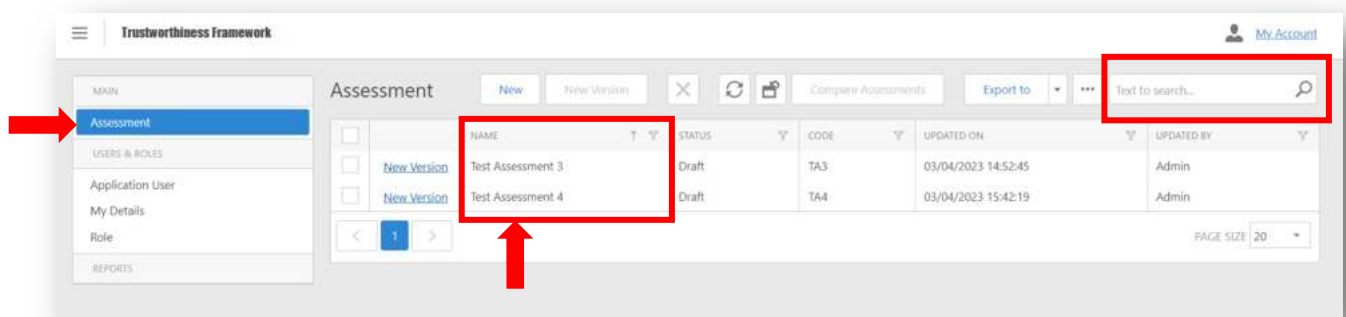


7. Click on the "**Save**" button, top right, to save the changes.
8. The flag on the "**Force Password Change**" box confirms the success of the operation, and force the user to change the password at the first log-in.

5. Creating, searching and viewing an assessment

In order to view assessments, follow the steps:

1. Select the "**Assessment**" page from the left menu.
2. Search by entering your request in the "**Search**" box at the top right.
3. Click on the name of the assessment to access the detail page.



In case the assessor needs to create a new assessment, there are two possibilities:

- a) By clicking the **New** button, a completely new assessment will be created, with all metrics having a zero score.
- b) By clicking the **New Version** link in the row of an existing assessment, it is possible to create a new version. That is, a new assessment which inherits the scores of the previous assessment, useful to assess a new version of the EIDS.

6. Editing an assessment

To speed up the assessment process, it is possible to automatically assign all metrics related to standards to which the system is already complaint. To do so, select the "**Standard**" tab by selecting the complaint box in the right column:

The screenshot shows the 'TrustworthinessFramework' web application. The 'Test Assessment 1' interface has three tabs: 'Assessment', 'Standards', and 'Pillars'. The 'Pillars' tab is active, displaying a table of pillars. The 'Security' pillar is highlighted in red. A radar chart on the right shows scores for various pillars.

	DESIGN SCORE	OPERATIONAL SCORE
NAME		
Ethics	0	0
Privacy	0	0
Reliability	0	0
Resiliency	0	0
Robustness	0	0
Security	0	0

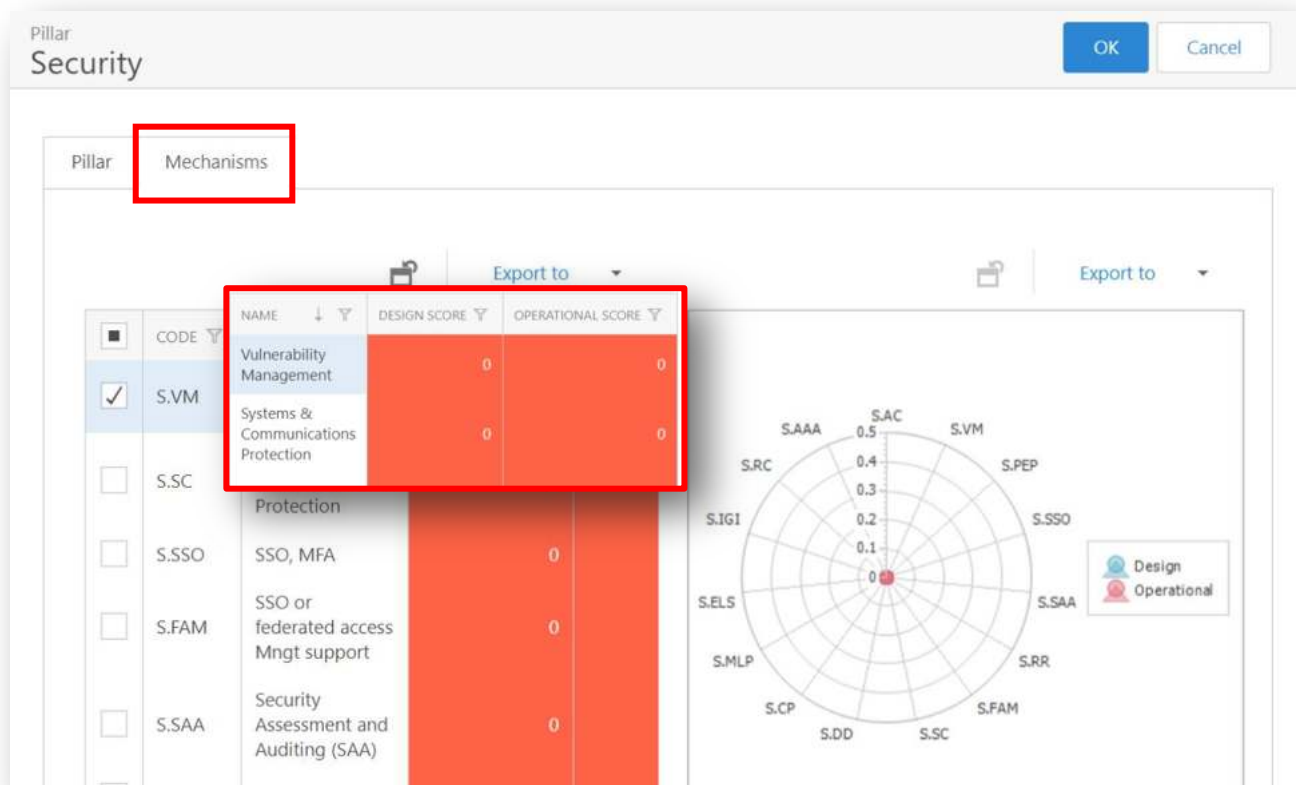
After having chosen the standards, it is possible to assess every specific pillar by selecting the **Pillars** tab and clicking on the row of the pillar to be modified.

The tab **Pillar** shows the design and operation scores, as well as the number of mechanisms.

The screenshot shows the 'Pillar Security' form. The 'Pillar' tab is selected, showing fields for CODE, NAME, NO MECHANISMS, DESIGN SCORE, and OPERATIONAL SCORE.

CODE:	DESIGN SCORE:
Security	0
NAME:	OPERATIONAL SCORE:
Security	0
NO MECHANISMS:	
15	

Clicking on the ***Mechanisms*** tab, it is possible to retrieve the list of underlying mechanism and the corresponding polar diagram.



By clicking on a mechanism, a new web page will open showing the details of such entry. Tab ***Mechanism*** shows the associated scores, as well as the weights and metrics. Here, it is possible to check the Boolean field ***Exclude from Assessment*** to declare that such mechanism will not be assessed and shall be ignored during the scoring process.

The screenshot shows the 'Mechanism Vulnerability Management' details page. The 'Mechanism' tab is selected and highlighted with a red box. The page displays various fields for the mechanism, including its code, name, number of metrics, and scores. A red box highlights the 'Exclude From Assessment' checkbox, which is currently unchecked. The description field provides a detailed explanation of vulnerability management.

Field	Value
CODE	S.VM
NAME	Vulnerability Management
NO METRICS	7
DESIGN SCORE	0
DESIGN WEIGHT	1
OPERATIONAL SCORE	0
OPERATIONAL WEIGHT	1
Exclude From Assessment	☐
DESCRIPTION	Vulnerability management is a process for identifying, evaluating, prioritizing, and mitigating vulnerabilities in an organization's IT infrastructure. Vulnerabilities are weaknesses or flaws in software, hardware, or systems that could be exploited by attackers to gain unauthorized access, steal data, or disrupt operations.

By selecting the **Design** and/or **Operational** tabs, the user can view all the corresponding metrics, of either Boolean (Y/N) or Percentage type. Using the pre-defined choices **Selected Design/Operational Choice** the assessor can select a common system configuration, quickly configuring the metric below. In addition, the assessor can also operate on each specific metric, by assigning a score to the columns Boolean Value and Percentage Value. To simplify the assignment, only one column is editable (with light blue background). Finally, clicking on the **Info** link, the user can view the description of such metrics as well as possible related standards.

The screenshot shows the 'Mechanism Vulnerability Management' interface. At the top, there are 'Mechanism', 'Design', and 'Operational' tabs. The 'Operational' tab is highlighted with a red box. Below the tabs, there is an 'OPERATIONAL QUESTION:' section with a text input field containing the question: 'Which of the following sentences best matches the operational efficacy of your system?'. Below this is a 'SELECTED OPERATIONAL CHOICE:' dropdown menu showing 'N/A'. To the right of the dropdown are 'Info', 'Export to', and a printer icon. Below these is a table with the following columns: NAME, CODE, METRIC TYPE, PHASE, BOOLEAN VALUE, PERCENTAGE VALUE, and WEIGHT. The table contains four rows of metrics. The 'PERCENTAGE VALUE' column is highlighted in light blue. At the bottom of the table, there are navigation buttons '< 1 >' and a 'PAGE SIZE' dropdown set to '20'. At the very bottom, there are 'Save changes' and 'Cancel changes' buttons.

	NAME	CODE	METRIC TYPE	PHASE	BOOLEAN VALUE	PERCENTAGE VALUE	WEIGHT
Info	Mean-time-to-detect	S.VM.O4	Percentage	Operational	☐	0	1
Info	Mean-time-to-resolve	S.VM.O5	Percentage	Operational	☐	0	1
Info	Patching management in place	S.VM.O2	Boolean	Operational	☐	0	1
Info	Residual vulnerability	S.VM.O1	Percentage	Operational	☐	0	1

The user can recursively click **OK** or **Cancel** to confirm or cancel the changes, and to return to the previous level (mechanism and then pillar).

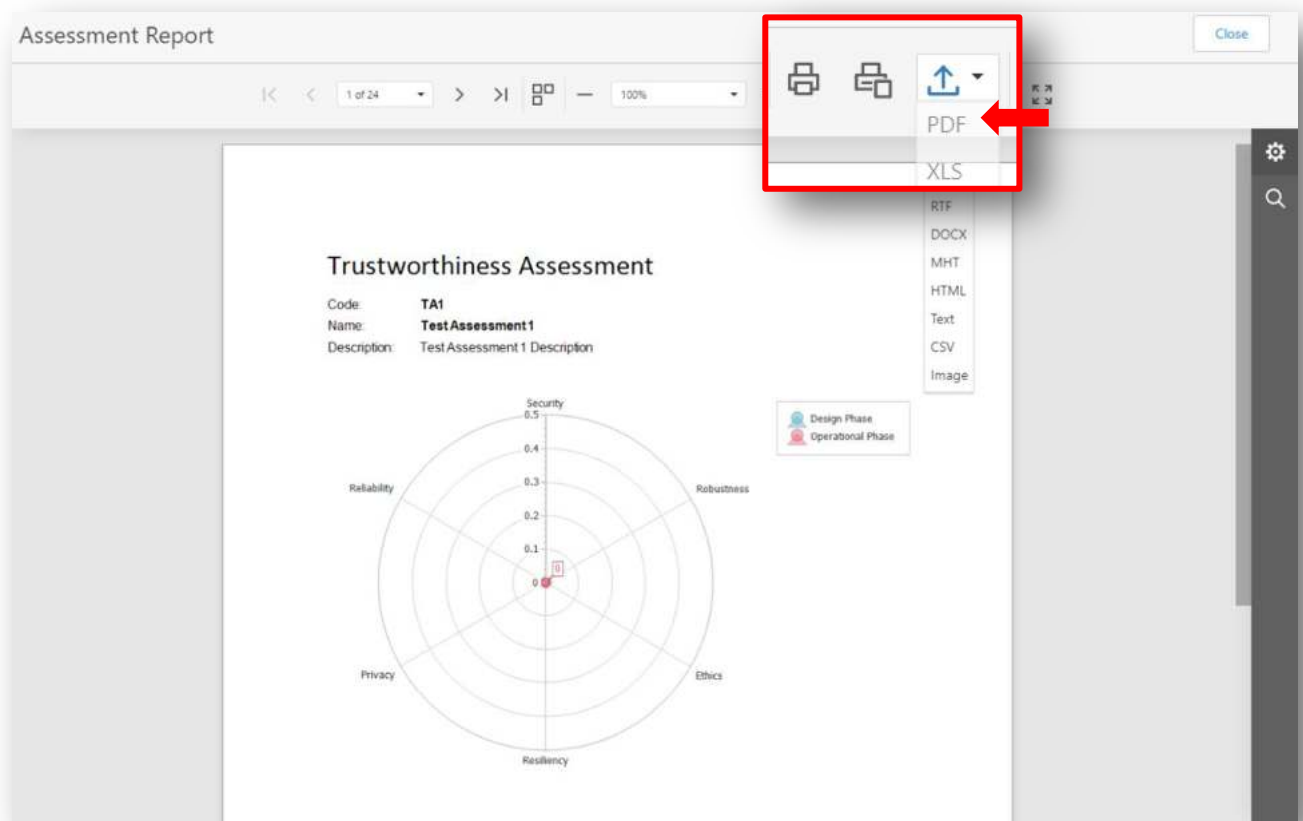
Note: Remember to save the assessment! However, if not properly saved, the tool will ask the user for confirmation before leaving the website.

7. Creating an assessment report

Assessments can be exported in a form of a report, following the steps:

- 1) Select the assessment in the **Assessment** section by clicking on its row.
- 2) In the detail page, click on the **Show Report** button.
- 3) A new webpage will open, containing the report. It is possible to print it directly, or to extract it in various formats (PDF, XLS, Word...)

In the assessment report, all levels (pillar-mechanism-metric) of the assessment are shown, using the same background colours used in the tool (see **Colour Legend** section below).



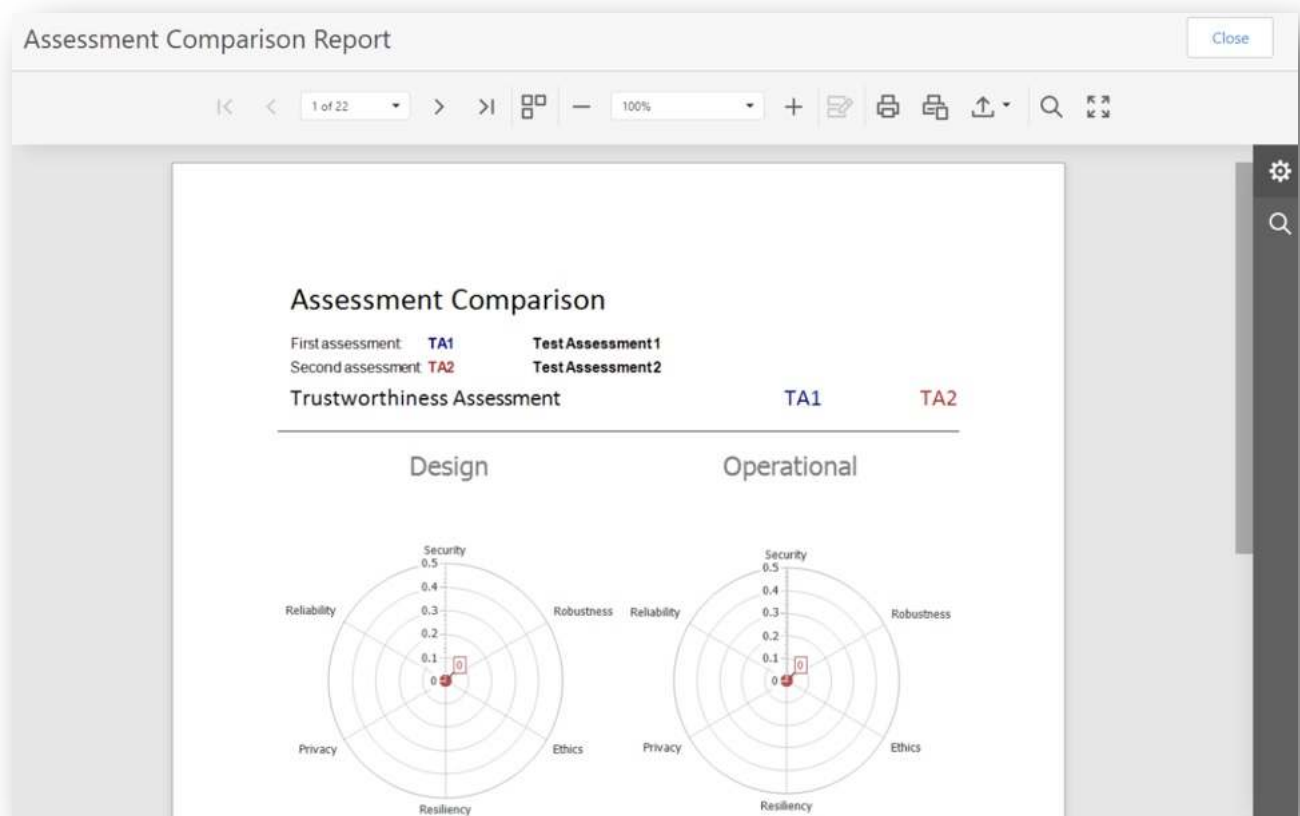
8. Creating a comparison report

Assessors (only) can generate a comparison report, between two existing assessments:

- 1) Select two assessments in the **Assessment** section by clicking on the corresponding checkbox at the beginning of the row.

- 2) When exactly two assessments are selected, the **Compare Assessments** button will be enabled. Clicking on the latter, will open a new webpage with the comparison report.
- 3) As in the case of the assessment report, the comparison can be exported in various formats.

In the comparison report, the two assessments are compared side by side, at all levels (pillar-mechanism-metric). The corresponding scores are elicited with a yellow background in case they differ.



9. Colour legend

In the assessment web pages, and in the assessment report, background colours are used to give a quick representation of the design and operational score of a pillar, of a mechanism and of a metric. Below the legend of such colours:

COLOUR	DESCRIPTION
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DEEP PINK	When applied to a metric, it means such metric is mandatory and it forces a cap at the score of the corresponding mechanism and/or pillar. When applied to a mechanism or a pillar, it means that one or more mandatory metrics are not satisfied.
TOMATO RED	When applied to a metric, a mechanism, and a pillar, it indicates that the corresponding score is $\leq 33\%$.
LEMON CHIFFON	When applied to a metric, a mechanism, and a pillar, it indicates that the corresponding score is between 33% and 66%.
LIGHT GREEN	When applied to a metric, a mechanism, and a pillar, it indicates that the corresponding score is $> 66\%$.
TRANSPARENT	Used at mechanism level, indicates that such mechanism has been excluded from the assessment.